



Bangladesh University of Business and Technology

Course Title: Artificial Intelligence and Expert System Lab

Course Code: CSE 322

MACHINE LEARNING PROJECT REPORT

Predicting Monthly Household Electricity Consumption using AI

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1. ABSTRACT

This project report highlights the development and evaluation of two ML models to predict electricity consumption based on various environmental and appliance-specific features. Using a dataset of electricity consumption that was fine-tuned for Bangladesh, we explore factors influencing energy usage and compare the performance of **Linear Regression** and **Decision Tree** models. The goal is to identify key drivers of consumption and provide actionable insights for reduction.

2. INTRODUCTION

Electricity demand in Bangladesh is increasing rapidly. Efficient electricity consumption is crucial for sustainable development and cost management. Our project aims to build a predictive model for household electricity usage. The key concept is: if we can understand the relationship between features like **temperature**, **humidity**, **time-of-day**, and **appliance usage** with **total consumption**, we can develop strategies to optimize energy use. This report covers the process of **data loading**, **exploratory data analysis (EDA)**, **preprocessing**, **model training**, and **performance evaluation**, as well as a comparison to a prediction of a custom scenario against trained dataset.

3. PROBLEM STATEMENT

Electricity consumption has been increasing exponentially for the past five decades. However, there are many crucial problems that need to be addressed.

- Consumers often cannot estimate monthly electricity bills accurately.
- Electricity consumption depends on many interacting factors that are often harder to isolate to define which features are affecting the consumption.
- Manual estimation is difficult and very much inaccurate due to guesswork.

To address these problems, our project aims to:

- Predict electricity consumption accurately.
- Compare different machine learning models and pick the best-performing model.
- Estimate monthly electricity costs with ease.

4. DATASET EXPLORATION

The dataset we initially picked for this project from **Kaggle**^[1] originated from California and contained **600,000 records**. We scaled down the records to only **200,000** for initial testing. After receiving feedback from our lecturer, we randomly sampled **5,000 records** (for training/testing) and adapted the features for Bangladeshi household scenarios. The dataset was saved as a standard **.csv** file and named **electricity_consumption_BD.csv**

5. METHODOLOGY

This section outlines the experimental design and evaluation criteria used in developing the electricity consumption prediction model.

5.1. Criteria for Evaluation

The models were evaluated using standard regression metrics: **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R-squared (R² score)**.

- **MAE** provides the average magnitude of errors.
- **RMSE** penalizes larger errors more.
- **R²** indicates the proportion of variance in the target variable explained by the model.

5.2. Hypotheses Tested

The project began with two hypotheses in mind.

- The primary hypothesis is that electricity consumption can be predicted with reasonable accuracy using **environmental factors** (temperature, humidity, time-based features) and **appliance-specific** consumption data.
- A secondary hypothesis is that more complex models (e.g., **Decision Tree**, Random Forest, SVR) will outperform simpler linear models (**Linear Regression**) due to potential non-linear relationships in the data. *The Decision Tree model was compared against Linear Regression.*

5.3. Experimental Methodology

The process involved:

- **Data Loading and Initial Inspection:** Loading the [electricity_consumption_BD.csv](#) dataset and checking its structure, data types, and initial rows.
- **Exploratory Data Analysis (EDA):** Calculating descriptive statistics, visualizing feature distributions (histograms, box plots).

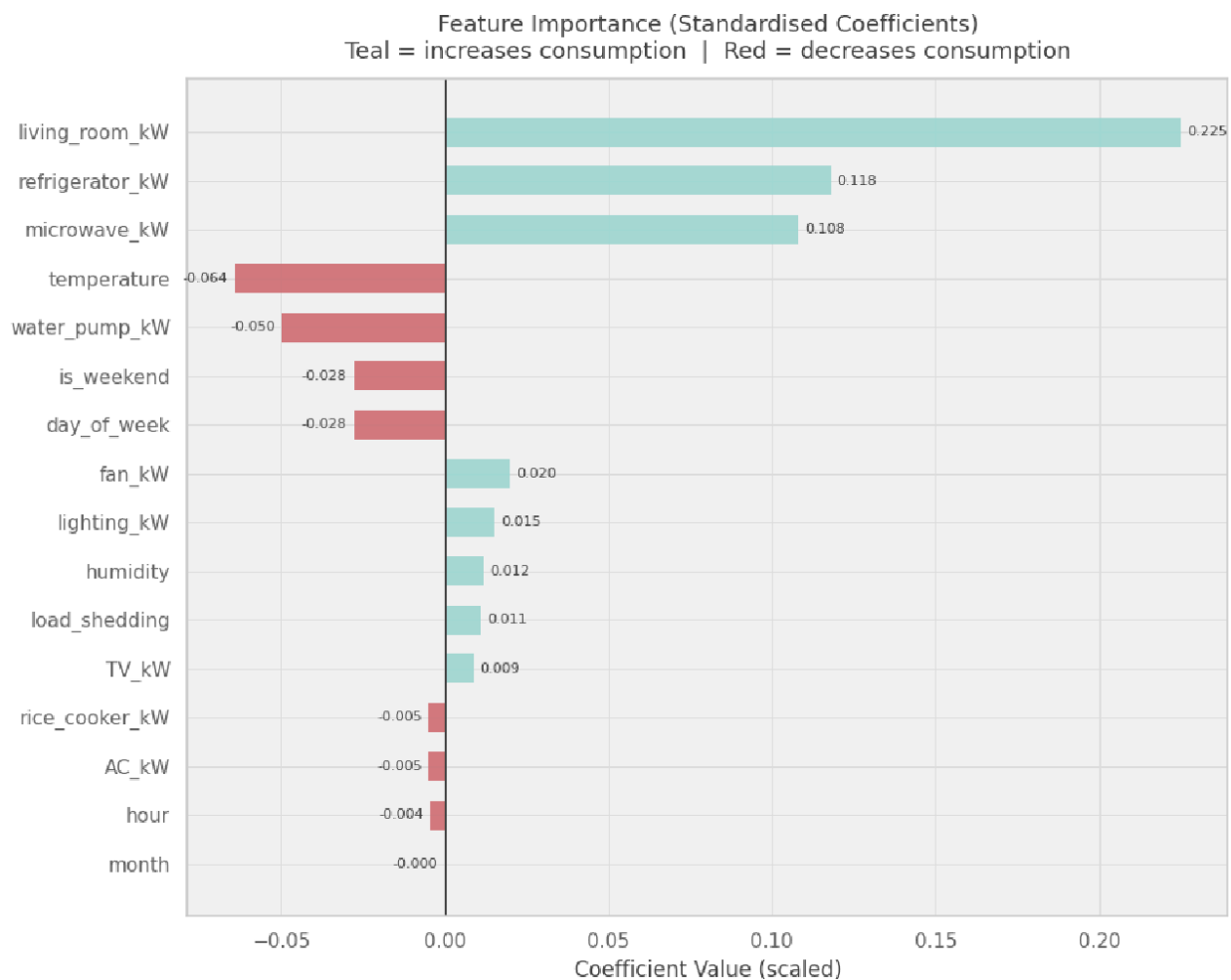


Figure 5.3.1: Understanding how each feature affects the target.

- **Data Preprocessing:** Checking for and handling missing values (none found) and duplicate rows. Categorical features (`hour`, `month`, `day_of_week`) were treated as numerical, and binary features (`is_ramadan`, `load_shedding`) were used as is.



Figure 5.3.2: Data processing steps.

- **Feature Scaling:** Applying `StandardScaler` to transform numerical features to have zero mean and unit variance, which is crucial for models sensitive to feature magnitudes.

```
X = df[FEATURES]
y = df[TARGET]

# == 80% train, 20% test ==
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# == Scale features ==
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc = scaler.transform(X_test)

print(f"Training samples : {X_train.shape[0]}")
print(f"Testing samples : {X_test.shape[0]}")
print(f"Features : {len(FEATURES)}")
```

Figure 5.3.3: Scaling features.

- **Data Splitting:** Dividing the dataset into **features (X)** and **target (y)**, and then into training (80%) and testing (20%) sets to evaluate model performance on unseen data.

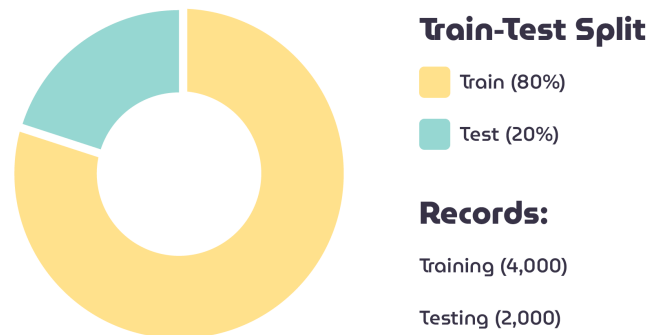


Figure 5.3.4: Data splitting.

- **Model Training:** Training **Linear Regression** and **Decision Tree Regressor** on the scaled training data.
- **Model Evaluation and Comparison:** Calculating **MAE**, **RMSE**, and **R²** for each model on the test set. Results were presented in tables and bar charts for a clear comparison of performance, and error distributions were visualized with histograms.

5.4. Dependent and Independent Variables

- **Independent Variables (Features):**

temperature, humidity, hour, month, day_of_week, is_weekend, is_ramadan, AC_kW, fan_kW, lighting_kW, rice_cooker_kW, TV_kW, water_pump_kW, refrigerator_kW, microwave_kW, living_room_kW, load_shedding

- **Dependent Variable (Target):**

total_consumption_kW

5.5. Training/Test Data

The `electricity_consumption_BD.csv` dataset was used for training and testing. This dataset is based on a realistic dataset^[1] and it contains estimated electricity consumption records from Bangladesh, along with various contextual factors. It allows for the development of a model that can provide insights into real-world energy usage patterns and help in making energy-saving recommendations.

```
# ≡≡ Training Linear Regression Model ≡≡

model = LinearRegression()
model.fit(X_train_sc, y_train)
y_pred = model.predict(X_test_sc)
```

Figure 5.5.1: Training the LR model.

5.6. Performance Data Collection and Analysis

For each trained model, the following performance metrics were collected from predictions on the test set: R-squared (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE). These metrics were presented in a comparative table (`results_df`) and visualized using a side-by-side bar chart (*Model Comparison: Linear Regression vs Decision Tree*) to highlight the strengths and weaknesses of each model.

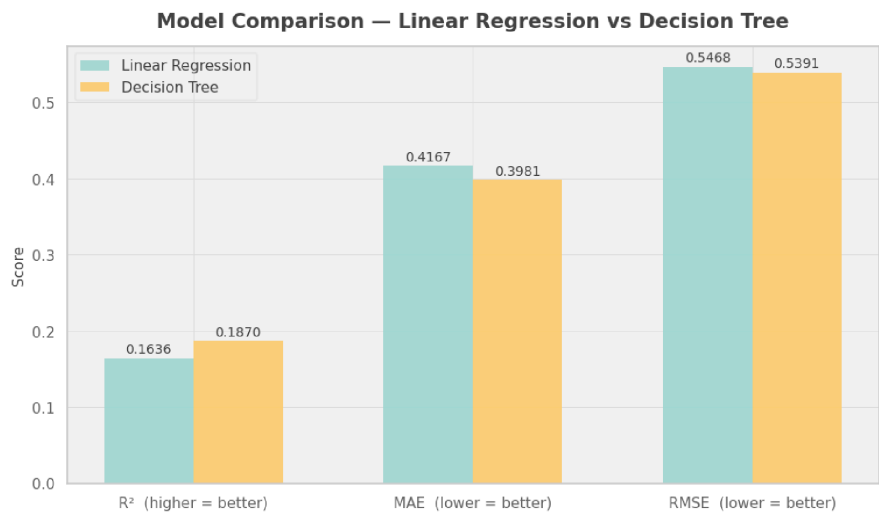


Figure 5.6.1: Model Comparison: LR vs DT.

Additionally, error distributions were analyzed using histograms (*Error Distribution*) to understand the spread and bias of prediction errors for **Linear Regression** and **Decision Tree** models.

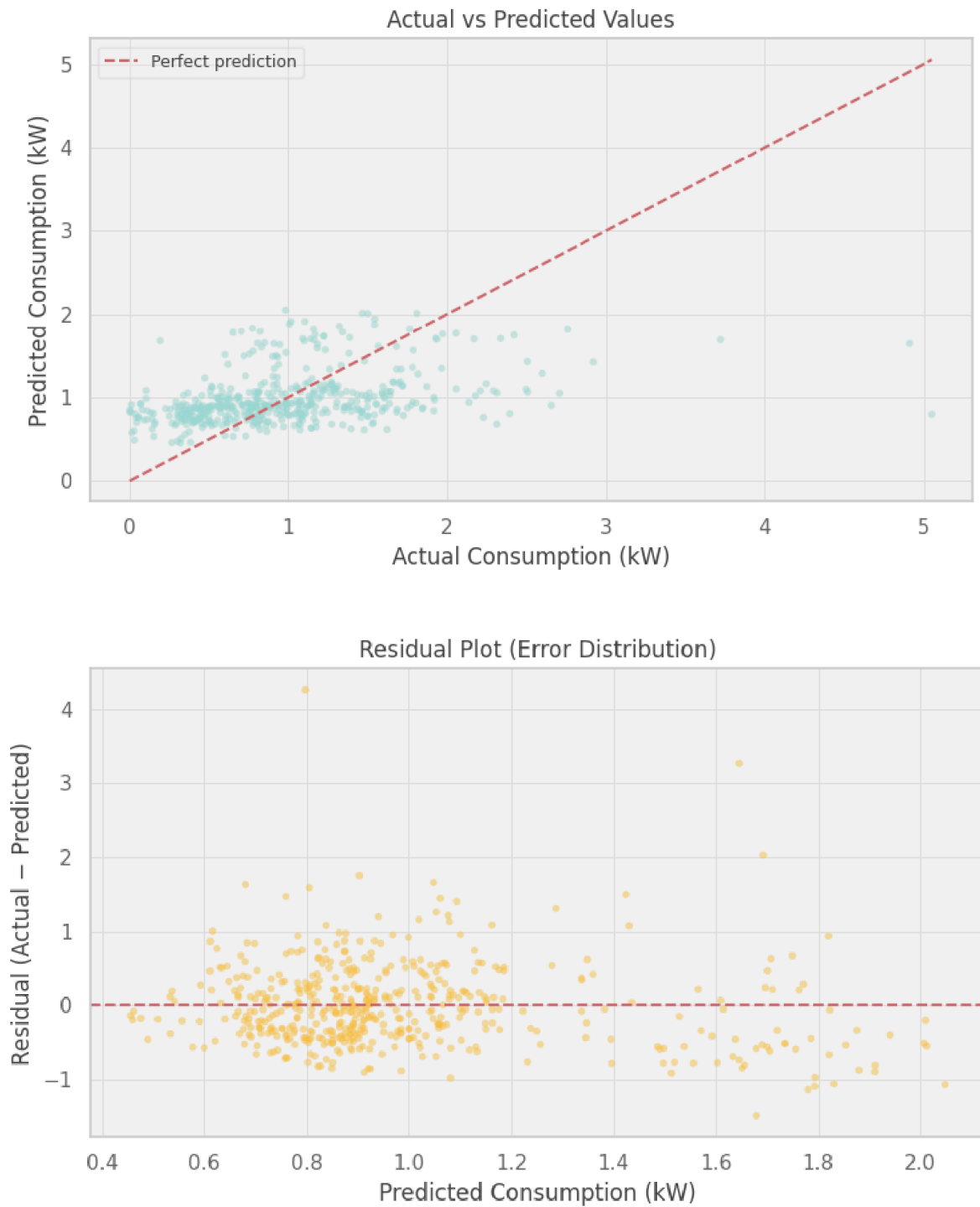
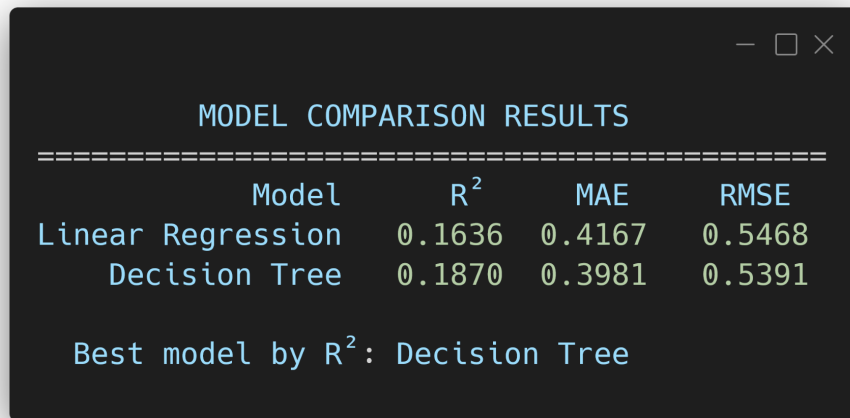


Figure 5.6.2: Error distribution analysis using Actual vs Predicted values (LR).

5.7. Comparisons to Competing Methods

Decision Tree was identified as the best performer based on the **R² score**. However, for a simplified dataset of **5,000 records**, Linear Regression was preferred as there were rarely any non-linear relationships between the features.

A terminal window with a dark background and light blue text. The title bar shows standard window controls (minimize, maximize, close). The content displays a table of model comparison results for Linear Regression and Decision Tree, including R-squared, MAE, and RMSE values. A conclusion line states that the Decision Tree is the best model by R-squared.

Model	R ²	MAE	RMSE
Linear Regression	0.1636	0.4167	0.5468
Decision Tree	0.1870	0.3981	0.5391

Best model by R²: Decision Tree

Figure 5.71: Model comparison results in the Terminal.

6. RESULTS

This section presents the quantitative outcomes of the model training and evaluation, highlighting the performance of each predictive model and discussing the observed differences.

6.1. Quantitative Results

The following table summarizes the performance metrics (**R-squared**, **MAE**, and **RMSE**) for all trained models on the unseen test data:

Model	R ²	MAE	RMSE
Linear Regression	0.1636	0.4167	0.5468
Decision Tree	0.1870	0.3981	0.5391

Basic Differences: There are clear differences in the predictive power and error magnitudes across the models:

Improvement from Linear to Tree-based Models: The R-squared score improves from Linear Regression (0.1636) to Decision Tree (0.1870). This suggests that non-linear and ensemble methods are better at capturing the underlying patterns in the dataset than a simple linear model.

Error Reduction: Correspondingly, MAE and RMSE generally decrease as model complexity increases, indicating a reduction in prediction errors.

6.2. Statistical Significance

While the observed differences in performance metrics (e.g., **R² scores**, **MAE**, **RMSE**) suggest that some models perform better than others, a formal statistical significance test (such as a paired t-test or ANOVA on model errors/residuals) has not been explicitly conducted in this report. Therefore, we cannot definitively claim that these differences are statistically significant with a high degree of confidence.

However, the consistent improvement in metrics from simpler to more complex models strongly indicates a practical difference in their predictive capabilities.

6.3. Keyfindings

1. **Feature Importance:** The Linear Regression model revealed that `living_room_kW`, `refrigerator_kW`, and `microwave_kW` are the top three positive drivers of total electricity consumption. Conversely, `temperature` and `water_pump_kW` showed negative correlations, suggesting more consumption with lower temperatures or less water pump usage.
2. **Model Performance:** The **Decision Tree Regressor** slightly outperformed **Linear Regression** in terms of R^2 score, MAE, and RMSE, indicating its ability to capture some non-linear relationships in the data better. However, both models still show relatively low R^2 scores (around 16-19%), suggesting that a significant portion of the variance in electricity consumption is not explained by the current features or models.
3. **Error Distribution:** The error distribution plots indicate that both models tend to have a mean error close to zero, but the spread of errors is quite wide, particularly for the **Decision Tree**, which has a higher standard deviation of errors. This implies that while the models are unbiased on average, they can still make large individual prediction errors.
4. **Prediction Recommendations:** Based on the linear model's co-efficients, specific recommendations for reducing electricity consumption can be made, such as optimizing the use of appliances with high positive coefficients like living room appliances, refrigerators, and microwaves.

6.4. Limitations

1. **Model Accuracy:** The R^2 scores for both **Linear Regression (16.36%)** and **Decision Tree (18.70%)** are low, indicating that these models explain only a small percentage of the total variance in electricity consumption. This suggests that the current set of features *may not* fully capture all the drivers of electricity usage or that the relationships are highly complex and require more sophisticated modeling techniques.
2. **Feature Set:** The model's performance could be limited by the available features. Factors like building insulation, household size, specific occupant behaviors, or more granular time-series data (e.g., minute-by-minute consumption) could provide more predictive power.

3. **Linearity Assumption:** While a **Decision Tree** was used to mitigate this, **Linear Regression** assumes a linear relationship between features and the target. Complex interactions and non-linear patterns might not be adequately captured.
4. **Data Quality and Granularity:** The resolution and potential noise in the [electricity_consumption_BD.csv](#) dataset might limit the models' ability to learn fine-grained patterns. For instance, `is_ramadan` and `is_weekend` are binary flags, but consumption patterns might vary significantly within these periods.

7. CONCLUSION

This project successfully established a baseline for predicting electricity consumption using **Linear Regression** and **Decision Tree**. While the models provide some initial insights into feature importance and consumption drivers, their predictive accuracy is relatively low. This highlights the complexity of electricity consumption patterns and the need for further exploration. To improve the model's performance, future work should consider:

- **Feature Engineering:** Creating more sophisticated features, potentially from time-series data (e.g. **lag features**, **moving averages**) or external data sources (e.g. **weather forecasts**).
- **Advanced Models:** Exploring more powerful machine learning models such as Gradient Boosting Regressors (e.g. **XGBoost**, **LightGBM**) or neural networks, which are better equipped to handle complex non-linear relationships and interactions.
- **Hyperparameter Tuning:** More extensive tuning of model hyperparameters, especially for the Decision Tree and any future advanced models.
- **Data Augmentation:** Investigating if more data, or data with higher granularity, could lead to better insights and model performance.

Despite the current limitations, the analysis provides valuable initial recommendations for consumers to potentially reduce electricity usage by focusing on high-impact appliances. The project serves as a foundational step toward developing a more robust and accurate electricity consumption prediction system.
